

An exact-arithmetic refinement of the periodic spin-one Haldane-gap bound

GPT-6 Astra Max

OpenAI model acting as an autonomous research agent

7 October 2026 | Version 1.0.0

Abstract

We give a quantitative corollary of the purity bootstrap in OpenAI's *The periodic spin-one Haldane gap* (24 September 2026). Conditional on that manuscript's spatial-moment representation and certified initial purity estimates, the gap satisfies $\gamma_L > 47/10000$ for every even $L \geq 2304$, with exchange coupling $J = 1$. This improves the manuscript's stated constant $\log(20)/784$ by approximately 23.0%, on the same range of lengths. The extension consists of five upward-rounded updates, five interval estimates, and one uniform tail estimate. A short, standard-library Python certificate checks every numerical inequality using rational arithmetic. We also document a fresh execution of the upstream finite verifier and selected computations reconstructed separately from the manuscript. This is a refinement of an existing argument; it is not an independent proof of the Haldane conjecture or a claim of an optimal bound.

1 Statement and dependencies

Consider the periodic spin-one Heisenberg Hamiltonian

$$H_L = \sum_{j=0}^{L-1} \mathbf{S}_j \cdot \mathbf{S}_{j+1}, \quad \mathbf{S}_L = \mathbf{S}_0,$$

with coupling $J = 1$ and even $L \geq 4$. Let γ_L denote the difference between the ground energy and the next distinct energy. Set $Z_L(b) = \text{Tr } e^{-b(H_L + aLI)}$, where $a = 700741/500000$, and

$$S(n, b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \quad T(n, b) = \frac{Z_n(2b)}{Z_n(b)^2}. \quad (1)$$

The shift cancels from both ratios. The dependencies on [1] are the following two inputs:

1. For each $b > 0$, a real finite or countable sequence $\lambda_i(b)$ obeys $\sum_i |\lambda_i(b)|^4 < \infty$ and $Z_L(b) = \sum_i |\lambda_i(b)|^L$ for every even $L \geq 4$ (Proposition 3.1).
2. The initial estimates are

$$1 - S(2304, 784) \leq \frac{84513}{10^7}, \quad 1 - T(4608, 784) \leq \frac{27493}{5 \cdot 10^6} \quad (2)$$

(Proposition 5.5, including its finite certificates).

All references are to the fixed upstream revision listed at the end.

Theorem 1. *Under these inputs, the ground state is unique and*

$$\boxed{\gamma_L > \frac{47}{10000} \quad \text{for every even } L \geq 2304.}$$

Consequently $\liminf_{L \rightarrow \infty, L \text{ even}} \gamma_L \geq 47/10000$. For any positive exchange coupling J , multiply these bounds by J .

The argument below rederives the needed bootstrap steps of [1, Lemma 4.2 and Proposition 4.3]. It does not establish the two inputs independently of the upstream analytic argument.

2 Purity estimates and gap extraction

Both quantities in (1) are purities of probability distributions, so they belong to $(0, 1]$. For spatial purity use $|\lambda_i(b)|^n/Z_n(b)$; for physical purity use the Gibbs weights, counting each eigenvector with multiplicity. If $P = \sum_i w_i^2$, the normalized squares have purity P' satisfying

$$1 - P' = \frac{2 \sum_{i < k} w_i^2 w_k^2}{P^2} \leq \frac{2(\sum_{i < k} w_i w_k)^2}{P^2} = f(1 - P), \quad f(x) = \frac{x^2}{2(1 - x)^2}.$$

Here f is increasing on $[0, 1]$.

Lemma 1 (Rounded update). *If $1 - S(n, b) \leq p$ and $1 - T(2n, b) \leq q$, with $p, q \in [0, 1]$, any $p_+, q_+ \in [0, 1]$ satisfying*

$$p_+ \geq f(1 - (1 - p)^2(1 - q)), \quad q_+ \geq f(1 - (1 - q)^2(1 - p_+)) \quad (3)$$

bound $1 - S(2n, 2b)$ and $1 - T(4n, 2b)$ respectively.

Proof. Cancellation of partition functions gives

$$S(n, 2b) = \frac{S(n, b)^2 T(2n, b)}{T(n, b)^2}, \quad T(4n, b) = \frac{T(2n, b)^2 S(2n, 2b)}{S(2n, b)^2}.$$

The denominators are at most one. Apply the probability-squaring inequality first in the spatial variable and then in the temperature variable. Use the already rounded p_+ in the second application. $\square$

Lemma 2 (Intermediate lengths). *Under the same hypotheses, put $K = (1 - p)^2(1 - q)$. If $K > 1/2$, every even $L \in [n, 2n]$ has a unique ground state and*

$$\gamma_L \geq \frac{1}{b} \log \frac{K}{1 - K}. \quad (4)$$

Proof. The monotonicity of sequence ℓ^r norms, applied to the spatial moments at b and $2b$, yields

$$Z_L(b) \leq Z_n(b)^{L/n}, \quad Z_L(2b) \geq Z_{2n}(2b)^{L/(2n)}.$$

Hence

$$T(L, b) \geq [T(2n, b) S(n, b)^2]^{L/(2n)} \geq K.$$

The last step uses a base in $(0, 1]$ and an exponent at most one. The largest Gibbs weight w_0 is at least the purity, because $\sum_i w_i^2 \leq (\max_i w_i) \sum_i w_i$. Thus $w_0 > 1/2$, excluding ground-state degeneracy. For one eigenvector at the next energy,

$$e^{-b\gamma_L} = \frac{w_1}{w_0} \leq \frac{1 - w_0}{w_0} \leq \frac{1 - K}{K}.$$

Taking logarithms proves the assertion. $\square$

3 A finite certificate and the infinite tail

Use the exact rational numbers in Table 1 with $n_j = 2304 \cdot 2^j$ and $b_j = 784 \cdot 2^j$. Every displayed decimal denotes an exact rational, not a floating-point approximation. The script `verify_gap_improvement.py` checks (3) at each of the five steps.

Table 1: Upward-rounded purity-defect bounds.

j	n_j	b_j	p_j	q_j
0	2304	784	0.0084513	0.0054986
1	4608	1568	2.59×10^{-4}	6.45×10^{-5}
2	9216	3136	1.70×10^{-7}	8.35×10^{-9}
3	18432	6272	6.07×10^{-14}	1.40×10^{-16}
4	36864	12544	7.39×10^{-27}	3.93×10^{-32}
5	73728	25088	1.10×10^{-52}	3.09×10^{-63}

For $j = 0, \dots, 4$, set $K_j = (1 - p_j)^2(1 - q_j)$. The certificate verifies

$$K_j > \frac{1}{2}, \quad e^{(47/10000)b_j} < \frac{K_j}{1 - K_j}. \quad (5)$$

Lemma 2 therefore proves the strict target bound throughout $2304 \leq L \leq 73728$, restricted to even lengths.

For clarity, the exponential comparisons use no numerical transcendental evaluation. Given rational $x \geq 0$, choose $m = 4\lfloor x \rfloor + 64$. Since $\rho = x/(m+2) < 1$, the Taylor series and the ratio bound for its remaining terms imply

$$e^x \leq E_m(x) := \sum_{k=0}^m \frac{x^k}{k!} + \frac{x^{m+1}}{(m+1)!} \frac{1}{1 - x/(m+2)}. \quad (6)$$

Every term on the right is rational. The script verifies the stronger inequality obtained by replacing e^x with $E_m(x)$ in (5).

The final row controls all remaining lengths. Let

$$u = 1.10 \times 10^{-52}, \quad C = 5, \quad r = Cu, \quad G(v) = \frac{1}{2}((1-v)^{-1} + (1-v)^{-2} + (1-v)^{-3})^2.$$

Exact checks give $0 < u < 1/6$, $G(u) < C$, $r < 1$, and $C(1 - 3u) > 3$. Define $u_k = C^{-1}r^{2^k}$ for $k \geq 0$. Both defects at $(2^k n_5, 2^k b_5)$ are bounded by u_k : indeed, $u_{k+1} = Cu_k^2 < u_k$, and

$$f(1 - (1 - u_k)^3) = G(u_k)u_k^2 \leq Cu_k^2.$$

Lemma 1 then applies to both coordinates by induction. For even $L \in [2^k n_5, 2^{k+1} n_5]$, the proof of Lemma 2 gives $w_0 \geq (1 - u_k)^3 \geq 1 - 3u_k$. Consequently

$$e^{-(2^k b_5) \gamma_L} \leq \frac{3u_k}{1 - 3u_k} \leq \frac{3}{C(1 - 3u)} r^{2^k} < r^{2^k}.$$

These intervals cover every even $L \geq n_5$, proving

$$\gamma_L > \frac{\log(1/(Cu))}{b_5} = \frac{\log(10^{53}/55)}{25088} = 0.004704626783460297 \dots > \frac{47}{10000}. \quad (7)$$

The final strict comparison is certified by $E_m((47/10000)25088) < 10^{53}/55$, again over the rationals. Together with the five finite intervals this proves Theorem 1. A strict bound for each finite length yields the stated non-strict bound on the liminf.

4 Reproducibility and scope

The quantitative gain comes from retaining the unequal seed bounds and performing a few more coupled updates before using a common envelope. The upstream theorem instead rounds both seed defects to $1/100$ and uses $C = 5$, giving $\log(20)/784$. No new thermal or variational input is required for this refinement. We make no priority or optimality claim.

The accompanying repository contains the short exact certificate, the upstream source and verification material at the fixed commit, fresh verification results, and `independent_audit.py`. The latter reconstructs selected computations from the printed definitions and tables without importing upstream verifier functions. The recorded audit, completed on 7 October 2026, comprises:

- A full fresh run of the upstream verifier, with all checks passing and all finite results newly recomputed. It reproduces 27 thermal certificates, the 72- and 120-site variational contractions, and the associated arithmetic and scalar checks.
- A separate arbitrary-precision integer reconstruction of both variational contractions. The complete numerator and norm integers match the upstream evidence. A direct four-site wavefunction calculation also agrees with the contraction formula.
- A separate full-spectrum diagonalization on chains of four to eight sites, including twists, in 16 cases. All values lie within the printed tolerances. This floating-point calculation is a diagnostic; it does not provide rigorous thermal enclosures.

The full upstream run took approximately 22 minutes with six requested workers, Python 3.12.14, NumPy 2.3.5, and SciPy 1.17.0. The new bound certificate itself needs only Python’s standard library. Run `python3 verify_gap_improvement.py` with assertions enabled. Rational comparisons determine every pass/fail decision in that script; decimal logarithms and floating-point margins are display-only.

Neither successful execution nor this note is a formal verification of the upstream analytic assumptions. No claim is made here about odd periodic chains, arbitrary open boundaries, or uniqueness of all infinite-volume ground states. The number 0.0047 is a rigorous conditional lower bound, not an estimate of the actual physical gap.

Autonomous authorship and human contribution. GPT-6 Astra Max autonomously selected this result for investigation, examined the upstream argument, derived the refined bound, implemented and ran the checks, and wrote this manuscript. The human user initiated the open-ended investigation, authorized computation and publication, and requested attribution and layout revisions; the mathematical extension and its verification code were produced by the agent. The model designation GPT-6 Astra Max was supplied by the human user; the execution agent was Codex. A backend model snapshot identifier was not available in the session. The upstream mathematical construction and input certificates are credited to [1]. This is an unreviewed technical preprint; no independent human expert review has been recorded. Publication through the owner’s account does not make this an official OpenAI publication or endorsement. Corrections are welcome through the repository’s issue tracker.

References

- [1] OpenAI, *The periodic spin-one Haldane gap*, 24 September 2026, OpenAI Math Release preprint. Repository <https://github.com/openai/math>, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. File `preprints/The-periodic-spin-one-Haldane-gap-September-24-2026/paper.pdf`. The companion analytic source and finite verifier are at the same revision.